All India Villages SaaS Platform 🚀

A production-style Full Stack SaaS platform that provides a comprehensive REST API for India's village-level geographical data.

Built using:

- React.js
- Node.js
- Express.js
- PostgreSQL
- JWT Authentication
- Recharts
- Python ETL Pipeline

📌 Project Overview

This platform provides a centralized API infrastructure for accessing standardized Indian geographical hierarchy data:

Country → State → District → Subdistrict → Village

The system is designed for:

- B2B applications
- E-commerce platforms
- Logistics companies
- Address autocomplete systems
- Government data services

🚀 Key Features

✅ Authentication & Security

- User Registration
- User Login
- JWT Authentication

- Protected APIs
- API Key Generation
- Password Hashing using bcrypt

✅ Data Engineering

- ETL Pipeline using Python
- Excel Dataset Processing
- Automated Data Import
- PostgreSQL Normalized Database

✅ Search Engine

- Real-time Village Search
- Live Autocomplete
- Hierarchical Address Results
- Pagination Support

✅ Analytics Dashboard

- Total States
- Total Districts
- Total Subdistricts
- Total Villages
- Interactive Charts using Recharts

✅ SaaS Architecture

- REST APIs
- Modular Backend Structure
- Protected Routes
- Scalable Database Design

- API Infrastructure

🏗 Tech Stack

Technology	Usage
React.js	Frontend
Vite	Frontend Build Tool
Tailwind CSS	UI Styling
Node.js	Backend Runtime
Express.js	API Server
PostgreSQL	Database
Python	ETL/Data Processing
JWT	Authentication
bcryptjs	Password Security
UUID	API Key Generation
Recharts	Data Visualization

📁 Project Structure

```
Capstone Project
│
├── frontend
│   ├── src
│   │   ├── components
│   │   ├── services
│   │   └── App.jsx
│
├── backend
│   ├── controllers
│   ├── routes
│   ├── middleware
│   ├── db.js
│   └── server.js
│
├── data-engineering
│   ├── import_data.py
│   ├── inspect_dataset.py
│
└── README.md
```

🗄 Database Architecture

The platform uses a normalized PostgreSQL schema:

States
↓
Districts
↓
Subdistricts
↓
Villages

Additional tables:

- users
- api_keys

🔐 Authentication Flow

User Login
↓
JWT Token Generated
↓
Token Stored in LocalStorage
↓
Protected API Access

🔑 API Key Infrastructure

Authenticated users can generate API keys for consuming APIs securely.

Example:

POST /api/v1/generate-api-key

📊 Analytics Example

The platform currently supports:

- 28 States
- 510 Districts
- 5205 Subdistricts
- 537,611 Villages

🔍 Search API Example

GET /api/v1/search?q=ram&page=1

Example Response:


```
{
 "results": [
  {
   "village_name": "Rampur",
   "subdistrict_name": "Akkalkuwa",
   "district_name": "Nandurbar",
   "state_name": "MAHARASHTRA"
  }
```

]
}

🧠 ETL Pipeline

The ETL process:

- Reads Excel datasets
- Cleans raw government data
- Normalizes hierarchy
- Imports into PostgreSQL

Implemented using Python and Pandas.

⚡ Performance Optimizations

- Indexed Search Queries
- Pagination APIs
- Optimized SQL Queries
- Modular Backend Architecture

💻 Frontend Features

- Login/Register UI
- Protected Dashboard
- Real-time Search
- Analytics Charts
- Responsive Design

🚀 Setup Instructions

1️⃣ Clone Repository

git clone https://github.com/YuvrajHere2002/all-india-villages-saas-platform.git

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2️⃣ Backend Setup

cd backend
npm install
node server.js

3️⃣ Frontend Setup

cd frontend
npm install
npm run dev

4️⃣ PostgreSQL Setup

Create database:

CREATE DATABASE villages_db;

Import schema and datasets using ETL scripts.

📸 Screenshots

Dashboard

- Analytics Cards
- Charts
- Real-time Search

Authentication

- Login Page
- Register Page

APIs

- JWT Protected Routes
- API Key Generation

🧪 API Endpoints

Method	Endpoint	Description
POST	/register	Register User
POST	/login	Login User
GET	/analytics	Analytics Data
GET	/search	Village Search
POST	/generate-api-key	Generate API Key

🔮 Future Improvements

- Redis Caching
- Rate Limiting
- Admin Dashboard
- NeonDB Deployment
- Vercel Deployment
- Advanced Analytics
- API Usage Tracking

🎯 Learning Outcomes

This project demonstrates:

- Full Stack Development
- Database Engineering
- Authentication Systems
- SaaS Architecture
- REST API Design
- ETL/Data Engineering
- Real-world Problem Solving

👨‍💻 Author

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🏁 Final Note

This project was developed as a production-style capstone project simulating real-world SaaS platform engineering using large-scale Indian geographical datasets.